IOP Publishing Measurement Science and Technology
Meas. Sci. Technol. 35 (2024) 035011 (10pp) https://doi.org/10.1088/1361-6501/ad128d

A vibration correction system for cold atom gravimeter

Meiying Guo*, Jinhai Bai, Dong Hu, Zhimin Tang, Jianqi You, Runnan Chen and Yu Wang

Changcheng Institute of Metrology and Measurement, Beijing 100095, People's Republic of China

E-mail: 1152027430@qq.com

Received 24 July 2023, revised 16 November 2023
Accepted for publication 5 December 2023
Published 12 December 2023

Abstract

Cold atom gravimeters are used to measure the absolute value of the gravitational acceleration. To mitigate the influence of seismic noise on the mirror, a vibration correction system is used. A commercial seismometer CMG-3ESPC is employed to measure the mirror vibration. The transfer function between the mirror vibration and the output of the seismometer is regarded as a proportional element with a time delay. The best proportional element and the best time delay are obtained by minimizing the standard deviation of the fitting residuals of the interference fringe $\alpha - P$ with vibration correction. This system not only exhibits excellent environmental adaptability but also eliminates the need for rough estimation of local gravitational acceleration during the vibration correction process. The system was built on the homemade cold atom gravimeter to carry out experiments in the laboratory. Compared with the measurement results without vibration correction, the standard deviation of the gravity measurement results was reduced from 120.5 μGal to 4.2 μGal, resulting in a significant reduction by a factor of 28.7. Meanwhile, its sensitivity was significantly improved from 330.8 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$ to 18.0 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$ by performing vibration correction. In the future, this system is expected to carry out comprehensive verification experiments in more complex noisy environments.

Keywords: cold atom gravimeter, ground vibration, vibration correction, seismometer

1. Introduction

Cold atom gravimeters are used to measure the absolute value of gravitational acceleration (g, about $9.8\,\text{m}\,\text{s}^{-2}$) [1–4]. A gas of cold atoms are applied as the test mass. The cold atoms are released from the trap in a vacuum chamber and the gravitational acceleration is measured by a Mach–Zehnder atomic interferometer [5, 6]. In the interferometer, the Raman beams vertically enter the vacuum chamber from the top and are retro-reflected by a mirror at the bottom. The mirror is affected by vibration noise, which will couple into interferometric measurements. In a quiet environment, such as laboratories located in suburban with isolated foundations, vibration noise results in scatter of hundreds of μGal ($1\,\mu\text{Gal} = 10^{-8}\text{m}\,\text{s}^{-2}$) in gravity measurements, while in a noisy environment, such as laboratories located near subway, it leads to scatter of tens of mGal ($1\,\text{mGal} = 10^{-5}\text{m}\,\text{s}^{-2}$) in gravity measurements [7–10].

A vibration correction system is a way to mitigate the vibration noise in cold atom gravimeters. Research on the vibration correction system for absolute gravimeters has a long history both domestically and internationally [11–14]. One notable example is the atom gravimeter developed by LNE-SYRTE at Observatoire de Paris in France. It employs a combination of passive isolator and vibration correction system. A seismometer and a mirror are both placed on the platform of the passive isolator. The output of the seismometer and the atomic interference fringe signal are recorded simultaneously. To obtain the actual motion of the mirror, they study two types of filters for vibration correction. One is to employ a combination of the recursive infinite impulse response (IIR) filter and low-pass non causal filter to preprocess the output of the

* Author to whom any correspondence should be addressed.

© 2023 IOP Publishing Ltd

seismometer. The introduced phase shift caused by the motion of the mirror is calculated, and is used to correct the interference fringe [15–17]. This approach has increased the measurement sensitivity of absolute gravimeters from 50 μGal $\sqrt{\text{Hz}}^{-1}$ to 14 μGal $\sqrt{\text{Hz}}^{-1}$, achieving a threefold improvement. The other utilizes a pure delay to compensate for the phase shift of the seismometer. This is done by simply shifting the acquisition of the seismometer data by this delay [16]. The Leibniz Universität Hannover in Germany processes the output of the seismometer using digital low-pass and high-pass filters to calculate vibration phase shift and correct the interference fringe [18, 19]. This approach has also been proven to effectively improve the measurement sensitivity of absolute gravimeters from 440 μGal $\sqrt{\text{Hz}}^{-1}$ to 92 μGal $\sqrt{\text{Hz}}^{-1}$. Zhejiang University in China implemented a vibration correction system in its atom gravimeter. A commercial software is used to correct the measured transfer function of the seismometer, obtaining the actual motion of the mirror. Experimental results have demonstrated that it improved the measurement sensitivity of absolute gravimeters by 3–4 times [20, 21]. The National University of Defense Technology in China conducts a detailed analysis of the measurement noise introduced by vertical and horizontal coupling-induced vibration. They have also accurately calibrated the transfer function between the output of the accelerometer and the motion of the mirror. Results show that the application of vibration correction improves the resolution of gravity measurement from 37.27 μGal@144 s to 6.236 μGal@144 s, achieving a factor of 5.98 enhancement. University of Science and Technology of China employs a three dimensional vibration correction method [22]. They perform pre-processing on the three-dimensional vibration signal to remove high frequency noise and correct for distortions. The distortion correction was achieved using a first-order phase filter and two low-pass filters. Afterwards, they calculate the phase shift introduced by the corrected vibration signal, in order to obtain the atomic transition probability of the cold atom gravimeter without vibration noise. While these vibration correction methods are effective, they do have limitations in terms of environmental adaptability. National Institute of Metrology in China applied the matured vibration correction method, which has been extensively used in optical absolute gravimeters, to the atom gravimeter. In this method, the transfer function between the motion of the mirror and the output of the seismometer is regarded as a proportional element with a time delay [9, 10, 23]. By iteratively exploring different parameters to minimize the standard deviation or root mean square error (RMSE) of fitting residuals in interference fringes composed of phase shift $\Delta\phi$ and transition probability P of atom [8]. This method improves environmental adaptability. Results indicated that the standard deviation of gravity measurement results decreases from 51.5 μGal to 22 μGal, achieving a reduction of 2.3 times. Nevertheless, a rough estimation of gravitational acceleration through preliminary measurements is needed before applying this method for vibration correction in atom gravimeters.

We propose a vibration correction system to enhance the environmental adaptability and measurement sensitivity of cold atom gravimeter. An optimization goal has been proposed, eliminating the requirement for rough estimation of local gravitational acceleration and enabling vibration correction in a more efficient manner. The manuscript begins by introducing the principles and algorithm of the vibration correction system for the cold atom gravimeter. The theoretical simulation is then conducted. Finally, the vibration correction system is integrated into the homemade cold atom gravimeter in the laboratory. A comprehensive assessment is carried out to evaluate the effectiveness of vibration correction. This assessment takes into account various factors such as the fitting quality of the atomic interference fringe without and with vibration correction, as well as the dispersion and sensitivity of the gravity measurement results.

2. Principles

2.1. Principles of vibration correction system for cold atom gravimeter

The cold atom gravimeter achieves atomic interference through two-photon stimulated Raman transition [2, 5, 24]. As shown in figure 1, two counter-propagating laser beams are used to illuminate the atom, resulting in the formation of a pair of Raman lasers. A gas of cold atom freely falls in vacuum. A sequence of three Raman laser pulses of $\pi/2$-π-$\pi/2$ are applied to the atoms at times t_1, t_2, and t_3, respectively. Three Raman pulses respectively achieve the splitting, reflection, and splitting of the cold atoms, similar to the Mach–Zehnder interferometer. During the second splitting process, cold atoms with the same energy from different paths overlap, resulting in atom interference.

A probe beam of light is applied to the falling cold atoms, the transition probability of atoms P is measured,

$$P = \frac{1}{2}\left(A + B\cos\Delta\Phi\right) \tag{1}$$

where A and B are coefficients, and $\Delta\Phi$ represents the atomic interference phase, which can be expressed as a linear combination of the phase of the three Raman pulses:

$$\begin{aligned}\Delta\Phi &= \Phi\left(t_1\right) + \Phi\left(t_2\right) + \Phi\left(t_3\right)\\ &= \Phi\left(t_1\right) - 2\Phi\left(t_1 + T\right) + \Phi\left(t_1 + 2T\right)\end{aligned} \tag{2}$$

where $\Phi(t_i)$ represents the phase of the Raman pulse at time t_i, T is the time interval between adjacent Raman pulses. In the ideal case, the relationship between the atomic interference phase $\Delta\Phi_{\text{id}}$ and the gravitational acceleration g at the measurement place is given by:

$$\Delta\Phi_{\text{id}} = \left(k_{\text{eff}}g - \alpha\right)T^2 \tag{3}$$

where k_{eff} is effective wave vector of the Raman laser and α is the chirp rate of the Raman laser. Substituting equation (3)

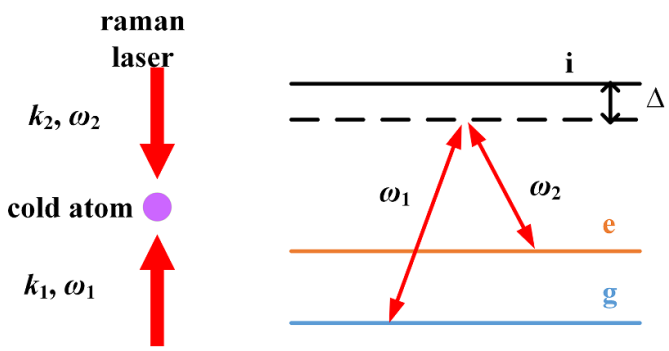

Figure 1. A stimulated Raman transition.

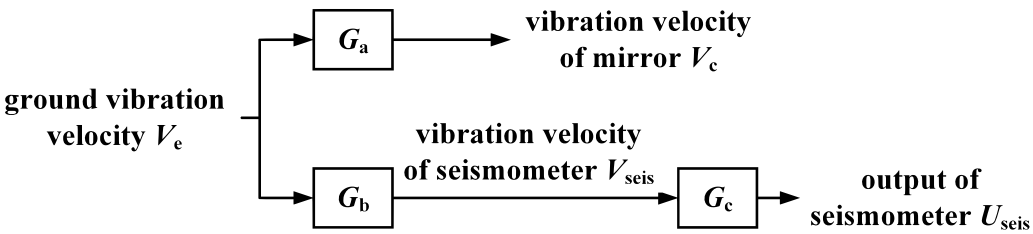

Figure 2. A diagram of the relationship between the output of the seismometer and the vibration velocity of the mirror.

into equation (1), the theoretical transition probability P_{id} can be obtained:

$$P_{id} = \frac{1}{2}\left(A + B\cos\left(\left(k_{eff}g - \alpha\right)T^2\right)\right). \tag{4}$$

By setting different values of time interval T, multiple sets of atomic interference fringe sequences $\alpha - P_{id}$ are obtained. According to equation (4), by identifying α^* corresponding to the overlapping region of the atomic interference fringe sequences at the valley or peak, the gravitational acceleration g can be obtained using the formula:

$$g = \frac{\alpha^*}{k_{eff}}. \tag{5}$$

The cold atom gravimeter utilizes a mirror to achieve two counter-propagating Raman lasers. In practical systems, the mirror is susceptible to ground vibration noise, which affects the phase of the Raman laser interaction with the cold atoms and causes a phase shift in the atomic interference fringe. Hence, the measured atomic interference phase $\Delta\Phi_m$ is expressed as:

$$\Delta\Phi_m = \left(k_{eff}g - \alpha\right)T^2 + \Delta\phi_{vib} + \Delta\phi_{other}. \tag{6}$$

Here, $\Delta\phi_{vib}$ represents the phase shift caused by the motion of the mirror and $\Delta\phi_{other}$ represents phase shift caused by other disturbances. $\Delta\phi_{vib}$ can be expressed as:

$$\begin{aligned}\Delta\phi_{vib} &= k_{eff}\left(y_c\left(t_1\right) - 2y_c\left(t_2\right) + y_c\left(t_3\right)\right)\\ &= k_{eff}\int_{-\infty}^{\infty} g_s\left(t\right)v_c\left(t\right)\mathrm{d}t\end{aligned} \tag{7}$$

where $y_c(t_i)$ represents the displacement of the mirror when the Raman pulse interacts with the cold atoms at time t_i, $v_c(t)$ is the velocity of the mirror at time t, and $g_s(t)$ is the sensitivity function of the cold atom gravimeter.

By combining equations (1)–(7), the relationship between the measured transition probability P_m and α in practical systems is given by:

$$\begin{cases} P_m = \frac{1}{2}\left(A + B\cos\left(\left(k_{eff}g - \alpha\right)T^2 + \Delta\phi_{vib} + \Delta\phi_{other}\right)\right)\\ \quad\;\; = \frac{1}{2}\left(A + B\cos\left(\left(k_{eff}g - \alpha_{cor}\right)T^2 + \Delta\phi_{other}\right)\right)\\ \alpha_{cor} = \alpha - \frac{\Delta\phi_{vib}}{T^2}. \end{cases} \tag{8}$$

From equation (8), it can be observed that when the main interference factor is vibration noise, the corrected chirp rate α_{cor} can be obtained by calculating the deviation of chirp rate caused by the vibration phase shift $\Delta\phi_{vib}$. This facilitates obtaining a new atomic interference fringe sequences $\alpha_{cor} - P_m$. By identifying α^*_{cor} corresponding to the overlapping region of the corrected atomic interference fringe sequences at the peaks or valleys for different T, the corrected gravitational acceleration g_{cor} can be obtained.

To obtain the vibration phase shift $\Delta\phi_{vib}$, as indicated by equation (7), it is necessary to acquire the actual vibration velocity of the mirror $v_c(t)$.

2.2. Vibration model of the mirror

A seismometer is used to measure the vibration velocity of the mirror. Both are placed on the ground. The relationship between the output of the seismometer U_{seis} and the vibration velocity of the mirror V_c is shown in figure 2. Here, G_a is the transfer function from the ground vibration velocity V_e to V_c,

G_b is the transfer function from V_e to the vibration velocity of seismometer V_{seis}, and G_c is the transfer function of the seismometer. Therefore, V_c can be expressed as:

$$V_c = \frac{U_{seis}}{G_c}\frac{G_a}{G_b} = H \cdot U_{seis} \tag{9}$$

Consider that within the bandwidth range, the amplitude-frequency characteristics of the seismometer are generally constant, and the phase-frequency characteristics are linear [25]. The transfer function G_c can be equivalently represented by a proportional element with a time delay within the bandwidth. Additionally, in a short period of time, the measurement environment does not undergo significant changes, so the ratio G_a/G_b can also be characterized by a proportional element with a time delay. Therefore, $v_c(t)$ can be expressed as:

$$v_c(t) = A \cdot v_{seis}(t+\tau) = A \cdot K_{seis} \cdot u_{seis}(t+\tau) \tag{10}$$

where A is the proportional element, τ is the time delay, and K_{seis} is the sensitivity of the seismometer.

When A and τ approach their theoretical values, the output of the seismometer $u_{seis}(t)$ can be used to obtain an approximation of the actual vibration velocity of the mirror $v_c(t)$. This, in turn, allows for the derivation of the approximate vibration phase shift $\Delta\phi_{vib}$ and corrected chirp rate α_{cor}, according to equations (7) and (8). In this case, the fitting effect of the atomic interference fringe $\alpha_{cor} - P_m$ is optimal. Hence, the optimization goal of vibration correction is to minimize the standard deviation of the fitting residuals of the corrected atomic interference fringes composed of chirp rate of Raman laser α and transition probability of atoms P. The vibration correction process does not require rough estimation of local gravitational acceleration.

2.3. Algorithm for vibration correction

The flowchart of the vibration correction algorithm for cold atom gravimeters based on the aforementioned principles is shown in figure 3.

3. Simulation

Before conducting experiments, it is essential to simulate the algorithm to validate its effectiveness. The simulation process is depicted in figure 4.

Assuming the theoretical gravitational acceleration g_0 is $9.8\,\mathrm{m\,s^{-2}}$, the time interval T is 170 ms, and the sampling rate of the vibration of mirror is 20 kHz. Five sets of interference fringe sequences are generated, each containing 20 pairs of $\alpha - P$. The vibration velocity $v_e(t)$ is generated based on the new low-noise and high-noise model (NLNM and NHNM) of global seismic background noise [26]. A maximum vibration phase shift introduced is approximately 11 rad. The theoretical time delay τ_0 and proportional element A_0 of the vibration

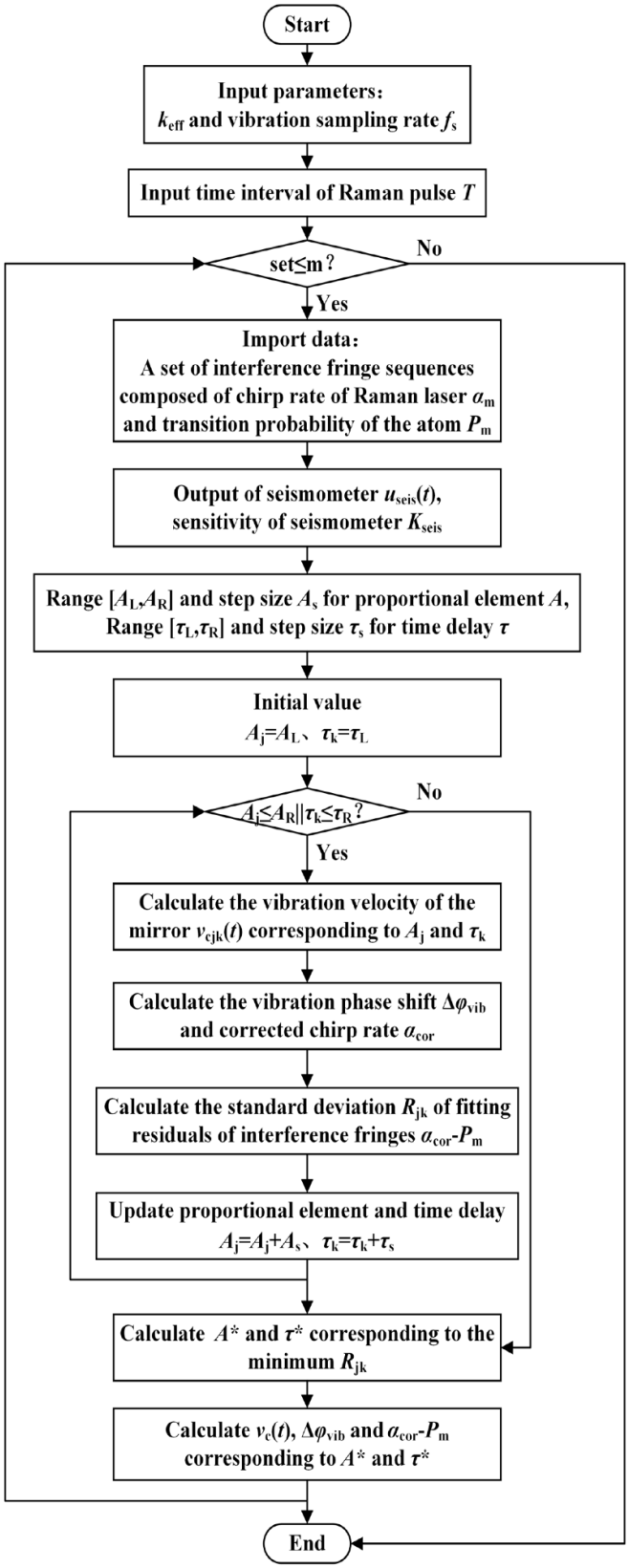

Figure 3. A flowchart for vibration correction algorithm used in cold atom gravimeters.

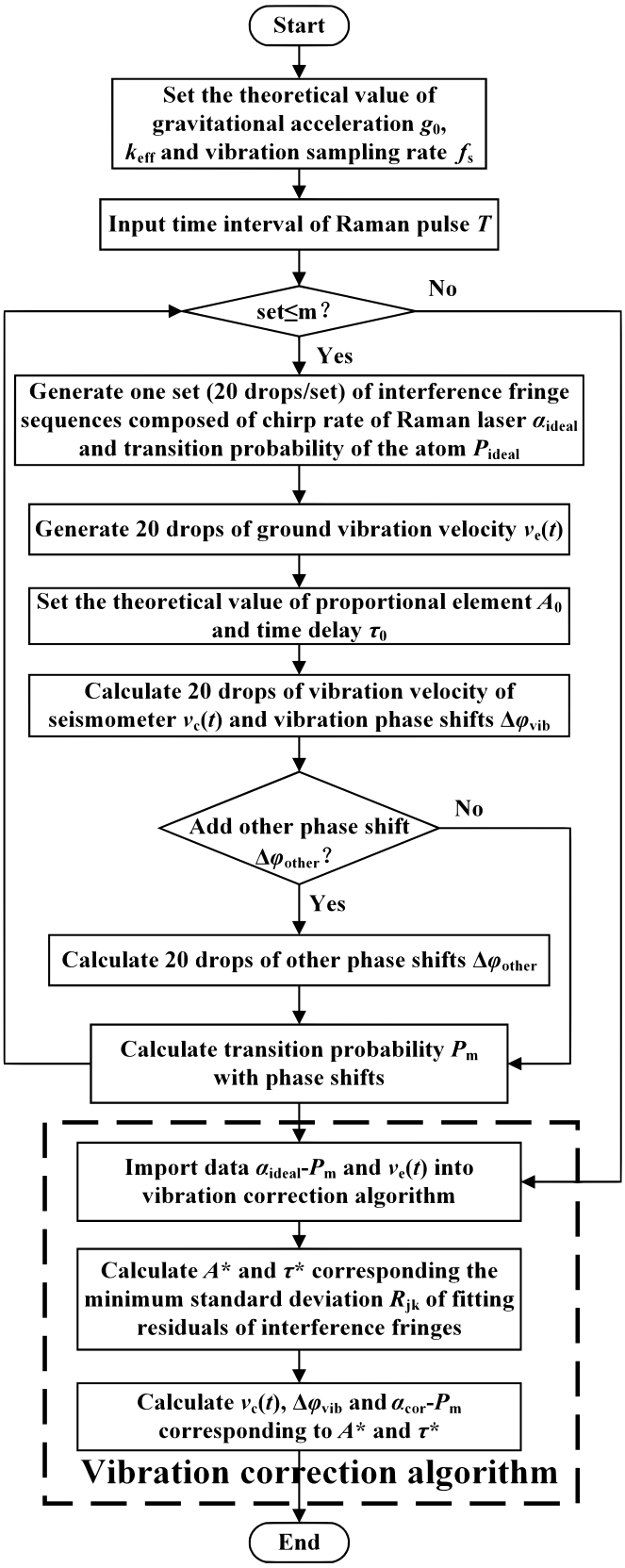

Figure 4. A flowchart of simulation algorithm.

is set to be 10 ms and 1.2. The generated 5 sets of interference fringe sequences $\alpha - P$ are shown in figure 5, where the brown dots represent the theoretical sequences, and the blue dots represent the sequences with introduced vibration phase shift.

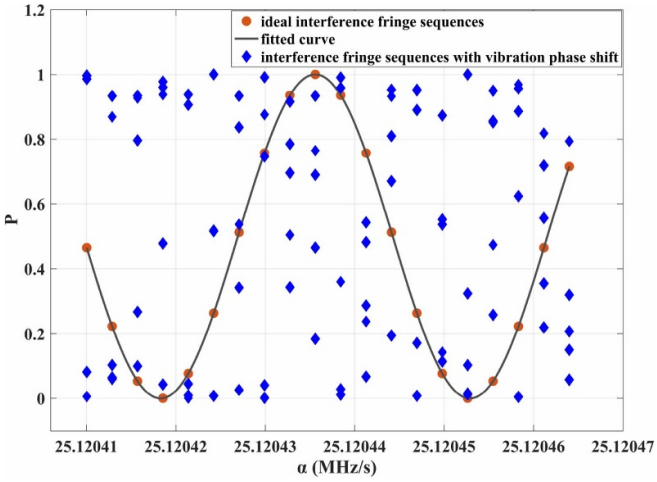

Figure 5. Theoretical interferometer fringes and interferometer fringes with introduced vibration noise.

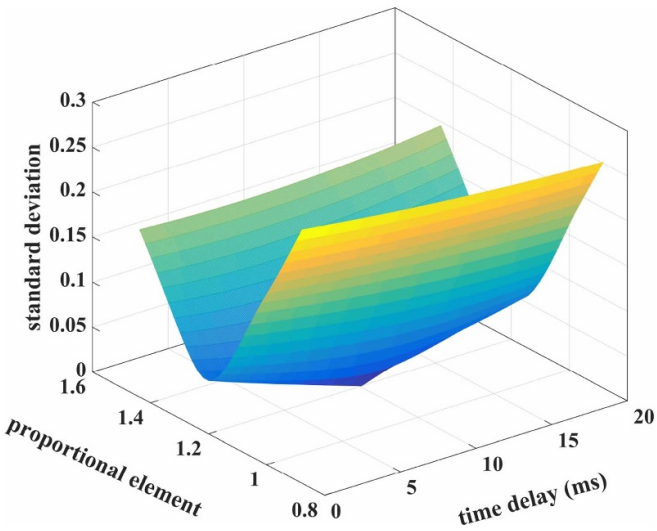

Figure 6. The standard deviation of the fitting residuals of simulated interference fringe with proportional element and time delay.

Five sets of interference fringe sequences $\alpha_{ideal} - P_m$ with vibration phase shift and the vibration velocity $v_e(t)$ are utilized as inputs to the vibration correction algorithm for analysis. As shown in figure 6, taking one set of interference fringe sequences as an example, when the standard deviation of the fitting residuals is minimized, the solved time delay τ^* is 10.10 ms, and the proportional element A^* is 1.20. The solved time delay and proportional element for the five sets of interference fringe sequences with vibration correction are shown in figure 7. The mean value of time delay τ^* is 10.07 ms with a standard deviation of 0.03 ms, and the mean value of proportional element A^* is 1.20 with a standard deviation of 0.00. These values are in close agreement with the theoretically parameters τ_0 and A_0. Figure 8 displays 5 sets of interference fringe sequences without and with vibration correction.

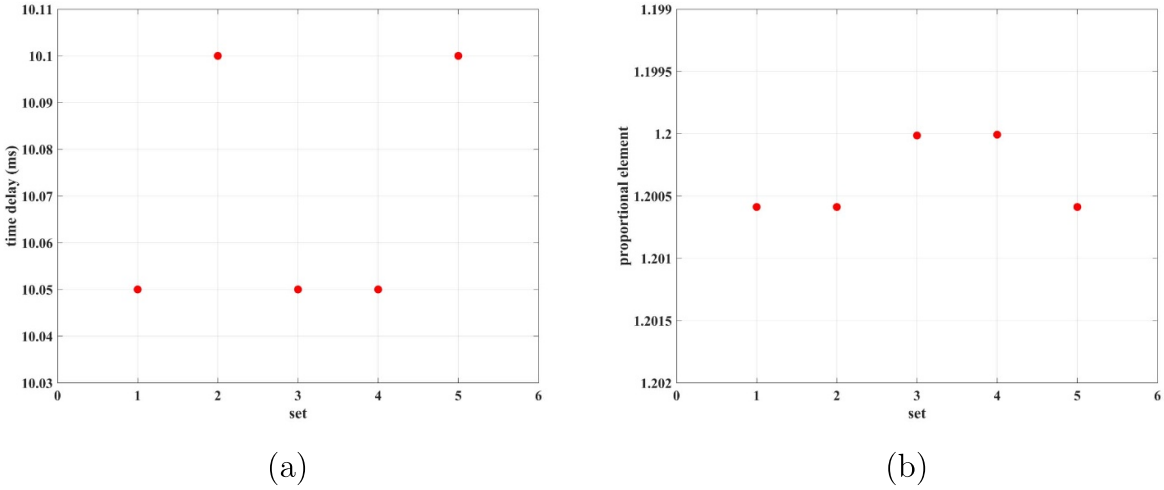

Figure 7. (a) Time delay, and (b) proportional element calculated by vibration correction algorithm.

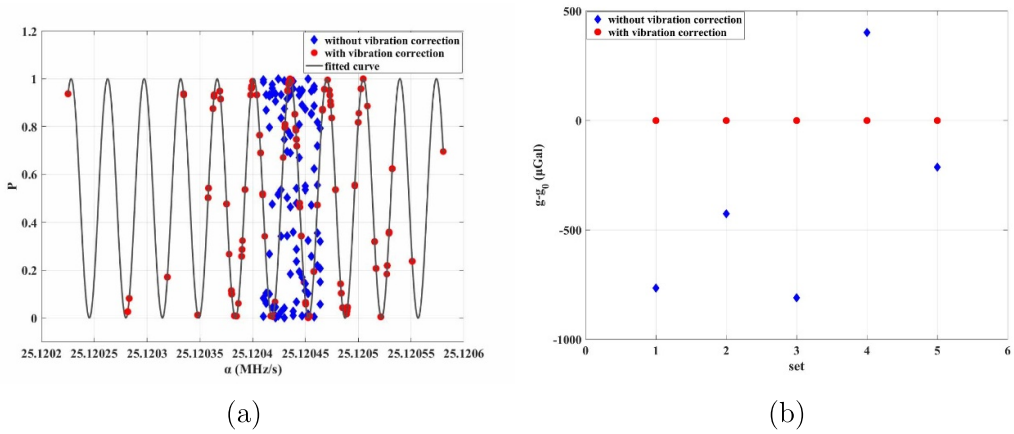

Figure 8. (a) Interference fringes $\alpha - P$ without and with vibration correction, (b) the variation of the measured g value without and with vibration correction.

It is observed that the corrected $\alpha_{\mathrm{cor}} - P_{\mathrm{m}}$ sequences align well with the fitting curve, and the variation between the solved g-values and the theoretical values g_0 decreases from $-362.5\,\mu$Gal to $-0.4\,\mu$Gal, with a standard deviation reduced from $493.2\,\mu$Gal to $0\,\mu$Gal. This verifies the effectiveness of the proposed algorithm.

4. Experiments

4.1. Experimental setup

The vibration correction system is integrated into the homemade cold atom gravimeter, as shown in figure 9(a). The effective wave vector of the Raman laser k_{eff} remains constant, and the time interval T is set at 250 ms. The chirp rate of the Raman laser α_{m} linearly increases. By measuring the transition probability of atoms P_{m}, the interference fringe sequences is obtained. The seismometer uses the CMG-3ESPC from Güralp Systems Ltd UK, as shown in figure 9(b) [25]. It has a sensitivity of 2000 $\mathrm{V\,(m\ s^{-1})^{-1}}$ and a bandwidth of 120 s–50 Hz. Its self-noise is lower than NLNM in the frequency range of 30 s–16 Hz. The output of the seismometer is collected using a 24-bit acquisition card, PCI-4462, from National Instruments.

4.2. Results and discussions

First, a set of interference fringe sequences is acquired, containing 20 pairs of $\alpha - P$. The relationship between the standard deviation of fitting residuals of interference fringes and two parameters : time delay and proportional element, with vibration correction is shown in figure 10(a). When the standard deviation of fitting residuals of interference fringes is minimized, the optimized time delay τ^* is 7.8 ms, and the proportional element A^* is 1.00. In figure 10(b), the blue and red dots represent the $\alpha - P$ without and with vibration correction, respectively. The results indicate that with vibration correction, the corrected fringes $\alpha_{\mathrm{cor}} - P_{\mathrm{m}}$ can align well with the fitting curve, demonstrating the effective suppression of phase shift introduced by ground vibration.

Vibration correction is carried out by utilizing 25 sets of fringe sequences. The optimal time delay and proportional

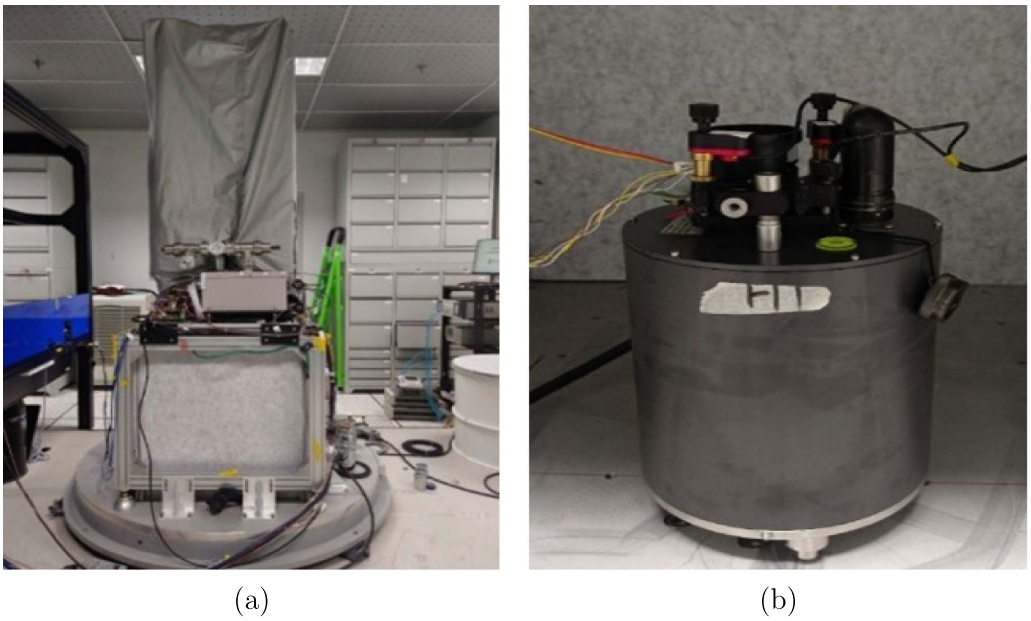

Figure 9. (a) The homemade cold atom gravimeter, (b) the vibration correction system built on the absolute gravimeter.

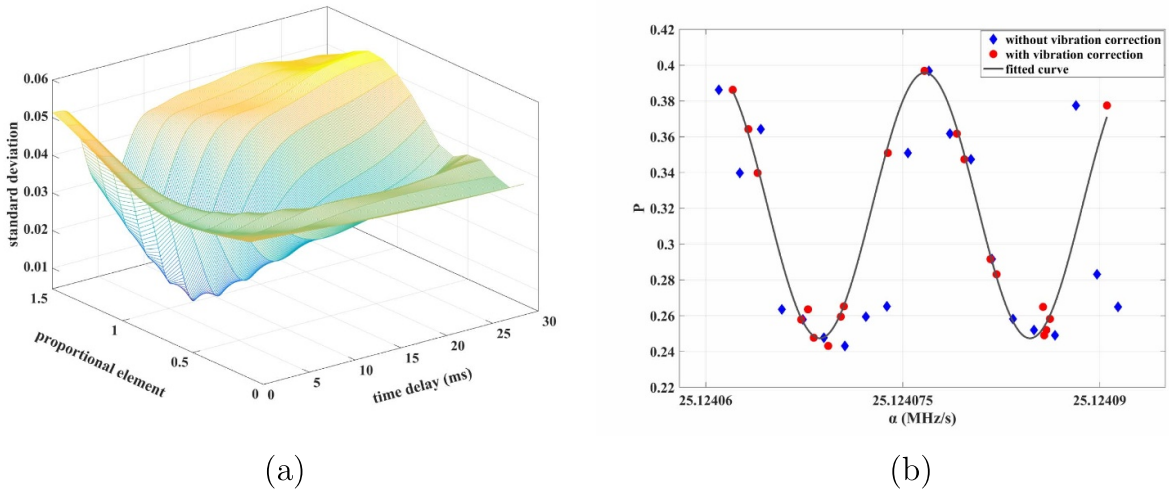

Figure 10. (a) The standard deviation of fitting residuals of measured interference fringe with proportional element and time delay, (b) the measured interference fringe $\alpha - P$ without and with vibration correction.

element for each set are illustrated in figures 11(a) and (b), respectively. It demonstrates this system has the potential to calculate optimal parameters in real time. The mean value of time delay τ^* is 8.3 ms with a standard deviation of 0.3 ms, and the mean value of proportional element A^* is 1.00 with a standard deviation of 0.04. The fringe sequences without and with vibration correction are displayed in figure 12. It is observed that the corrected fringe sequences $\alpha_{\mathrm{cor}} - P_{\mathrm{m}}$ exhibit a sinusoidal distribution, and the majority of the data points fall on the fitting curve. The standard deviation of the gravitational acceleration measurement results is reduced from 120.5 μGal to 4.2 μGal, a reduction of about 28.7 times, demonstrating that the vibration correction system can significantly attenuate the influence of ground vibration on absolute gravity measurements.

The vibration correction system is built on the cold atom gravimeter for a long-term observation. As shown in figure 13(a), the blue dots represent the gravity measurement results without vibration correction, the red dots represent the gravity measurement results with vibration correction, and the black curve represents the actual solid tides. The results show that the cold atom gravimeter, equipped with vibration correction, is capable of measuring solid tides. The gravity measurement results with vibration correction after eliminating the influence of solid tides are presented in figure 13(b). The significant fluctuations observed in the gravity measurement results with vibration correction at the 21st hour are indeed attributed to the occurrence of a 5.8 magnitude earthquake in the Off Sanriku of Japan at 6:48 am on 25 August, 2023.

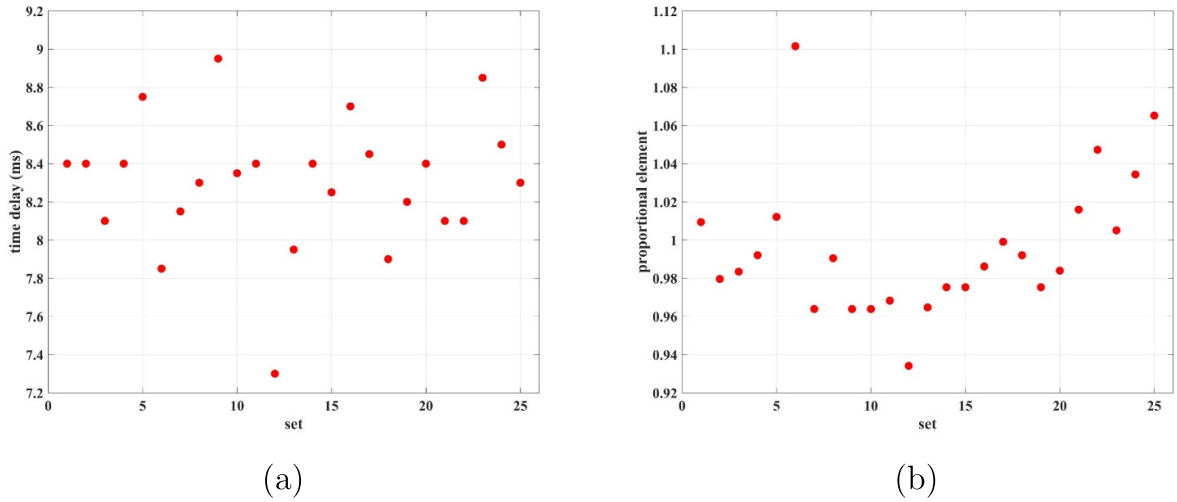

Figure 11. (a) Time delay, and (b) proportional element of 25 sets of interference fringes calculated by vibration correction algorithm.

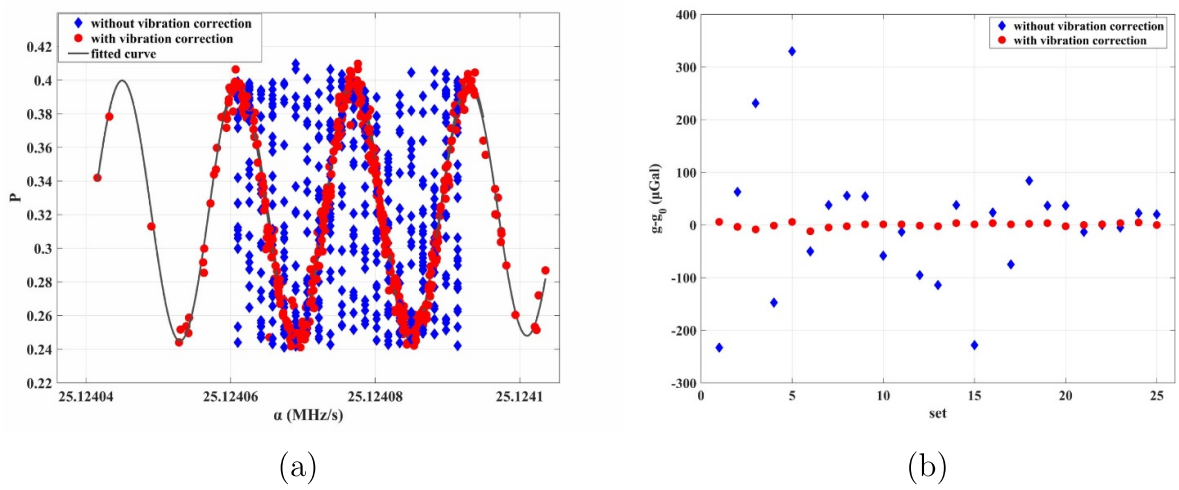

Figure 12. (a) 25 sets of interference fringes $\alpha - P$ without and with vibration correction, (b) the variation of the measured g value without and with vibration correction.

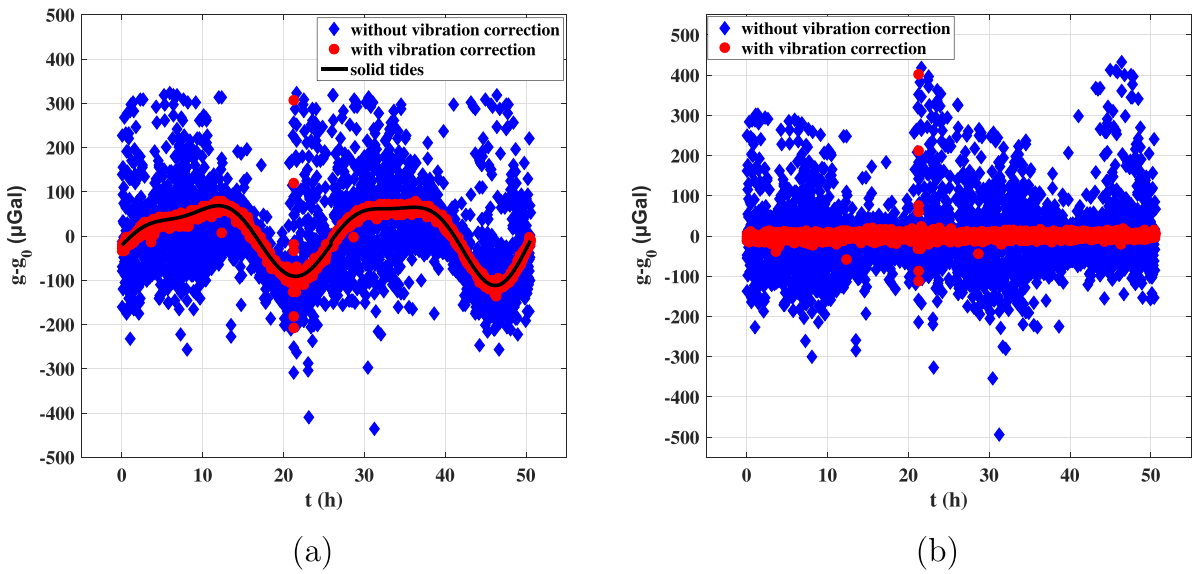

Figure 13. (a) The variation of the measured g value without and with vibration correction, (b) after eliminating the influence of solid tides.

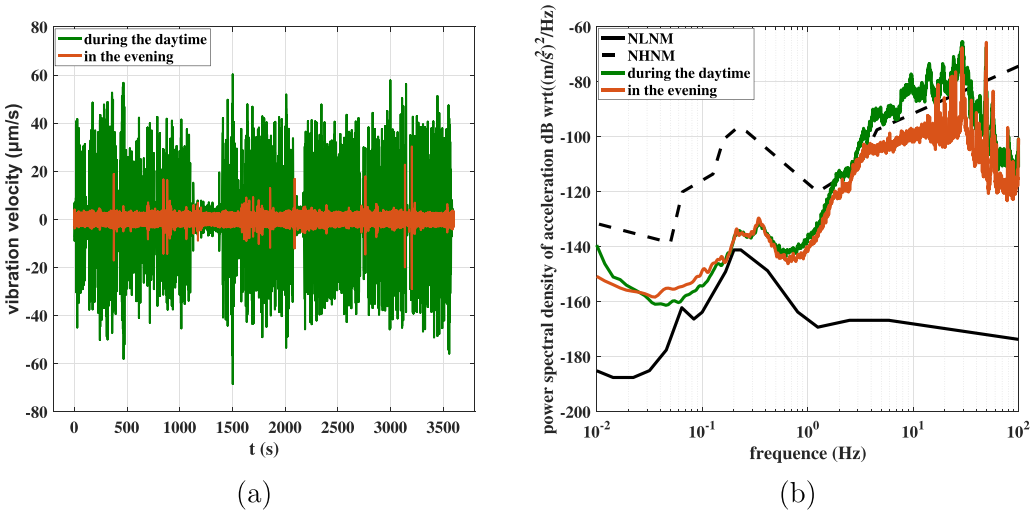

Figure 14. (a) Ground vibration velocity collected for one hour by seismometer during the daytime and in the evening, respectively, (b) The power spectral density of ground vibration acceleration noise.

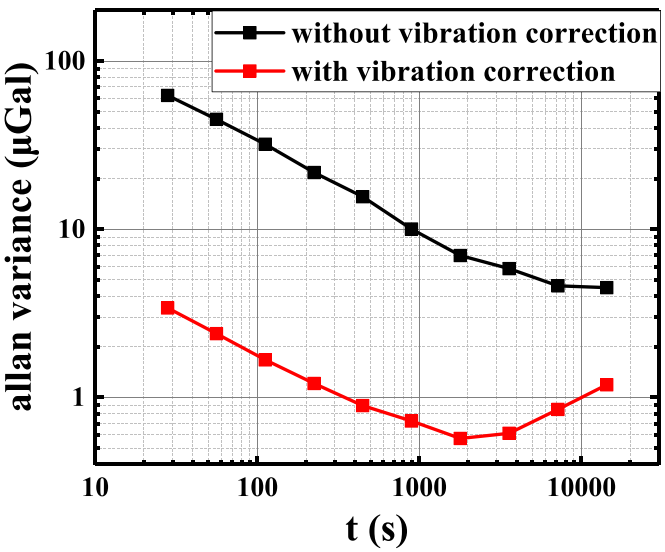

Figure 15. Allan variance of the gravity measurement results.

According to figure 13, it is observed that the dispersion of the gravity measurement results fluctuates, with larger variations occurring at certain times and smaller variations at other times when no vibration correction is applied. The probable explanations for this phenomenon are as follows. During the daytime, there was a higher level of personnel movement in the laboratory, resulting in increased environmental vibrations. The impact of these vibrations is shown in the green curve depicted in figure 14. In the evening, personnel had left, the testing environment became relatively quieter. This is depicted by the orang e curve in figure 14. Compared to the measurements during the daytime, the amplitude of ground vibrations decreased in the evening. Particularly for frequencies above 4 Hz, the power spectral density of the ground vibration was lower in the evening. However, as shown in figure 13(b), vibration correction effectively attenuates the influence of ground vibration on gravity measurements regardless of the environment. It indicates that the vibration correction system exhibits good adaptability to the environment and improves the precision of gravity measurements.

Figure 15 shows the Allan variance of the gravity measurement results without and with vibration correction. It illustrates that, with vibration correction, the sensitivity of gravity measurement results improves from 330.8 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$ to 18.0 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$, which is a remarkable enhancement by a factor of 18.4.

5. Conclusion

This paper presents a vibration correction system for cold atom gravimeters. The system consists of two components: a vibration sensor and an algorithm. The vibration sensor utilizes a velocity seismometer CMG-3ESPC. The algorithm regards the transfer function between the vibration velocity of the mirror and the output of the seismometer is a proportional element A with time delay τ. By iteratively adjusting the parameters A and τ, the system aims to minimize the standard deviation of the fitting residuals of the interference fringes $\alpha - P$. The system demonstrates excellent adaptability to environments and eliminate the requirement for rough estimation of local gravitational acceleration during the vibration correction process.

The vibration correction system is built on the homemade cold atom gravimeter. Experimental results demonstrate that this system effectively attenuates the impact of ground vibrations on absolute gravity measurements. The standard deviation of the gravity measurement results is reduced from 120.5 μGal to 4.2 μGal, a reduction of 28.7 times. Moreover, the sensitivity improves from 330.8 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$ to 18.0 $\mu\text{Gal}\sqrt{\text{Hz}}^{-1}$, demonstrating a remarkable factor of 18.4 enhancement. In the future, this system will be deployed on cold atom gravimeters to carry out comprehensive verification under different working conditions.

Data availability statement

The data cannot be made publicly available upon publication due to legal restrictions preventing unrestricted public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (Grant Nos. 61727819 and 12004360).

ORCID iD

Meiying Guo https://orcid.org/0000-0001-9624-3899

References

[1] Kasevich M and Chu S 1991 Atomic interferometry using stimulated Raman transitions *Phys. Rev. Lett.* **67** 181–4
[2] Peters A, Chung K Y and Chu S 2001 High-precision gravity measurements using atom interferometry *Metrologia* **38** 25–61
[3] Bodart Q, Merlet S, Malossi N, Pereira Dos Santos F, Bouyer P and Landragin A 2010 A cold atom pyramidal gravimeter with a single laser beam *Appl. Phys. Lett.* **96** 134101
[4] Farah T, Clairon A, Landragin A, Merlet S, Dos Santo F P and Louchet-Chauvet A 2011 Continuous g monitoring with atom interferometry *2011 Joint Conf. of the IEEE Int. Frequency Control and the European Frequency and Time Forum (FCS) Proc.* pp 1–4
[5] Zhou M K 2011 Experimental demonstration of an atom interferometry gravimeter *PhD Thesis* Huazhong University of Science and Technology
[6] Schmidt M 2011 A mobile high-precision gravimeter based on atom interferometry *PhD Thesis* Humboldt University of Berlin
[7] Chen B 2020 Development of portable atomic gravimeter and its vibration isolation system *PhD Thesis* University of Science and Technology of China
[8] Yao J M, Zhuang W, Feng J Y, Wang S K, Zhao Y, Wang S K, Wu S Q and Li T C 2022 A coefficient searching based vibration correction method *Acta Phys. Sin.* **71** 119101
[9] Wang G, Hu H, Wu K and Wang L J 2017 Correction of vibration for classical free-fall gravimeters with correlation-analysis *Meas. Sci. Technol.* **28** 035103
[10] Qian J, Wang G, Wu K and Wang L J 2018 A vibration correction method for free-fall absolute gravimeters *Meas. Sci. Technol.* **29** 025101
[11] Absolute quantum gravimeter (available at: www.muquans.com/product/absolute-quantum-gravimeter/)
[12] Canuteson E, Zumberge M and Hanson J 1997 An absolute method of vertical seismometer calibration by reference to a falling mass with application to the measurement of the gain *Bull. Seismol. Soc. Am.* **87** 484–93
[13] Brown J M, Niebauer T M and Klingele E 2001 Towards a dynamic absolute gravity system *Int. Association of Geodesy Symp.* vol 123 pp 223–8
[14] Lu H Y 2015 The optimization of the vibration isolation algorithms for a new generation of absolute gravimeter *PhD Thesis* Institute of Geophysics, China Earthquake Administration
[15] Le G J, Mehlstäubler T E, Kim J, Merlet S, Clairon A, Landragin A and Pereira Dos Santos F 2008 Limits to the sensitivity of a low noise compact atomic gravimeter *Appl. Phys.* B **92** 133–44
[16] Merlet S, Le G J, Bodar Q, Clairon A, Landragin A and Pereira Dos Santos F 2009 Operating an atom interferometer beyond its linear range *Metrologia* **46** 87–94
[17] Merlet S, Le G J, Bodart Q, Clairon A, Landragin A, Pereira Dos Santos F and Rouchon P 2010 Vibration rejection on atomic gravimeter signal using a seismometer *Gravity, Geoid and Earth Observation* (Springer) pp 115–21
[18] Richardson L L 2019 Inertial noise post-correction in atom interferometers measuring the local gravitational acceleration *PhD Thesis* Leibniz Universität Hannover
[19] Richardson L L *et al* 2020 Optomechanical resonator-enhanced atom interferometry *Commun. Phys.* **3** 208
[20] Xu A P, Wang Z Y, Wang Q Y, Fu Z J and Lin Q 2015 The restraint of the common-mode noise in the cold atom gravimeter *Geophys. Geochem. Explor.* **39** 58–62
[21] Xu A P 2015 The restraint of the common-mode noise in the atom interferometry gravimeter *PhD Thesis* Zhejiang University
[22] Chen B, Yang S J, Chen S and Pan J W 2018 *Vibration noise correction method for atomic interferometric gravimeters* China CN201811413684.2
[23] Wen Y, Wu K and Wang L J 2022 Analysis of vibration correction performance of vibration sensor for absolute gravity measurement *Acta Phys. Sin.* **71** 049101
[24] Wu B, Wang X L, Wang H L, Wang Z Y and Lin Q 2015 The current situation and tread of the gravimeter based on cold atom interferometer *Navig. Control* **14** 1–9
[25] Güralp Systems Limited 2016 CMG-3ESP operator's guide Güralp systems limited (available at: https://www.guralp.com/documents/html/MAN-ESP-0001/s1.htm)
[26] Peterson J 1993 Observations and modeling of seismic background noise *Open File-Report* 93-322 (U.S. Department of Interior Geological Survey)